**4** The probability distribution of the discrete random variable $X$ is given in Fig. 4.

| $r$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(X = r)$ | 0.2 | 0.15 | 0.3 | $k$ | 0.25 |

**Fig. 4**

**(i)** Find the value of $k$. **[2]**

$X_1$ and $X_2$ are two independent values of $X$.

**(ii)** Find $P(X_1 + X_2 = 6)$. **[3]**

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| **4(i)** | |
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| **4(ii)** | |
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